Mid Semester Examination- EVEN SEMESTER 2024- 2025

Name of the Program: B.Tech. Biotechnology — Semester: **4th**
Name of the Course: Bioprocess Engineering — *Course Code:* TBT 403

Time: 90 min — Maximum Marks: 50

SET-1

Note:
(i) Answer **all the questions** by choosing ***any one of the sub questions.***
(ii) Each question carries 10 marks

Q1	(10 marks)	CO2
(a)	Derive an expression for Malthus law and Double Reciprocal Plot. What is the significance of Double reciprocal plot?	
	OR	
(b)	Plot Batch Growth curve highlighting various phases in it. Define cryptic growth and Diauxic growth.	
Q2	(10 marks)	CO1
(a)	Highlight the relationship between del factor and sterilization time by deriving the Design Equation of sterilization.	
	OR	
(b)	Define Del factor. What are different methods of sterilization?	
Q3	(10 marks)	CO2
(a)	Apply concepts of mass balance to prove relationship between μ (specific growth rate) and D (Dilution rate) in CSTR.	
	OR	
(b)	Plot variation of Product with Time for 3 Kind of products obtained in Batch culture. Further, give expression of q_p in each case.	
Q4	(10 marks)	CO1
(a)	If the initial number of microbes is 10^8 and the Final number of microbes is 10^2, calculate the value of del factor. Explain Richards method of Del calculation.	
	OR	
(b)	Sketch the diagram of the continuous plate heat exchanger and discuss how it works.	
Q5	(10 marks)	CO2
(a)	Following data is observed while growing *Bacillus subtilis* in Batch mode: Time (hrs) 2 ,4, 6 ,8 ,10, 12 ,14 Biomass (g/l) 0.05 ,0.08, 1.4 ,1.6, 2.2, 2.8, 3.2 Substrate (g/l) 20, 16, 14, 11, 8, 5, 3 Calculate: 1.Overall specific growth rate 2.Doubling time of microbes	

	3. Yield of Biomass on substrate in time span 6 hrs to 12 hrs.	
	OR	
(b)	*Xanthomonas campestris* is being cultivated in a chemostat with a working volume of 500 mL. The feed stream has a substrate concentration (glucose) of 20 g/L, and the flow rate is 250 mL/h. The microorganism follows Monod kinetics with a μmax of 0.6 h^{-1} and a Ks of 0.5 g/L. The yield coefficient (Yx/s) for biomass production from the substrate is 0.4 g biomass/g substrate. **Calculate:** 1. The dilution rate (D). 2. The steady-state substrate concentration (S). 3. The steady-state biomass concentration (X). 4. The volumetric productivity (Q) of Chemostat.	